Nexia Tutor: An AI-Powered Language Personalized Learning System for Kids with Dyslexia and Reading Challenges

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***Abstract*—This research introduces Nexia Tutor, an AI-driven, microservices-based personalized language learning system designed for children with dyslexia. The system comprises three main services: a screening service for dyslexia identification, a report generation service for customized feedback, and an educational, game-based tutoring service tailored to individual needs. The screening service utilizes Random Forest algorithms, achieving an accuracy of 89.2%, to effectively diagnose learning difficulties. Personalized reports analyze dyslexia subtypes, identifying specific weaknesses and providing actionable guidance for parents. The tutoring service, powered by the BERT model, customizes games that target cognitive skills, such as visual memory and phonological awareness, and integrates Google Voice technology for auditory reinforcement. Findings indicate that the system's personalized approach enhances critical language skills in dyslexic children. Assessment data demonstrate improvements in comprehension, retention, and phonological awareness among users. The system's use of Natural Language Processing (NLP) and image generation technologies, such as BERT and DALL-E, further enriches the learning experience, making it interactive and accessible. Overall, this research underscores the potential of Nexia Tutor to offer timely, individualized interventions, aiding children with dyslexia in overcoming specific language challenges**

Keywords: Dyslexia, ITS, Personalized Tutoring, NLP, BERT, Random Forest

I. Introduction

Reading disorders significantly impact individuals, with early language delays, speech challenges, and language difficulties strongly linked to reading disabilities [1]. Children with dyslexia face unique challenges beyond reading, including learning difficulties, hyperactivity, and compromised psychosocial functioning, affecting productivity and well-being in educational and workplace settings [2]. These individual challenges vary widely, with specific deficits in areas such as phonological processing, visual processing, working memory, and attentional control, all of which require tailored intervention approaches.

Targeted interventions can address specific difficulties and enhance related skills. For instance, phonics-based interventions have significantly improved reading skills in students with intellectual disabilities by strengthening phonological awareness [3], while visual-attentional interventions have enhanced reading abilities in both dyslexic and neurotypical children by targeting visual processing skills [4]. However, current interventions often lack personalization and regular progress tracking, leading parents to seek assistance from centers that may overlook the educational needs of children with dyslexia [5].

Language learning challenges impact reading, spelling, and memorization for individuals with dyslexia and other learning disabilities. These reading disorders frequently co-occur with additional language, attention, and learning disorders, underscoring the crucial role of oral language skills in literacy development [6]. Without targeted support, deficits in these areas can lead to broader educational and social difficulties, ultimately affecting productivity in both academic and workplace settings. Thus, early intervention is essential for improved developmental outcomes [7].

Addressing these complex needs in children with learning disabilities (LD) requires pinpointing individual areas of difficulty, such as phonological processing deficits, which interfere with sound recognition, or visual processing challenges that impact reading fluency. Comprehensive screening combined with personalized tutoring can effectively tackle these specific challenges, ensuring a more productive and adaptive learning experience [8, 9]. Despite the prevalence of language learning applications, there remains a lack of platforms that integrate comprehensive screening with tailored language learning services capable of addressing the nuanced needs of individuals.

AI systems have demonstrated potential in dyslexia diagnosis, using tools like eye tracking and machine learning with high predictive accuracy [10, 11]. However, these systems often provide only binary classifications, which lack the granularity required to inform individualized interventions. Similarly, while AI-driven tutoring advancements have led to game-based educational experiences, these approaches frequently lack the specificity needed to target individual cognitive and learning challenges in dyslexia, thus limiting their effectiveness for personalized interventions [12].

Our solution bridges these gaps by integrating comprehensive screening with personalized tutoring. This approach advances through stages like categorizing specific learning difficulties, generating detailed PDF reports, implementing keyword extraction using BERT transformers, and creating

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educational games tailored to individual needs. By addressing the individual challenges faced by children with dyslexia, our contributions include a game-based screening service, sophisticated report generation, keyword extraction for enhanced textual analysis, and personalized educational games that cater to each child's unique learning profile. Data and functions traverse our system via message brokers and gateway services, facilitated by a centralized database, ensuring a seamless and responsive user experience.

II. Related Work

A. Large Language Models in Education

Large Language Models (LLMs) are increasingly used in educational tools to process language and provide personalized feedback. For example, Amira is an AI-driven system offering reading tutoring, oral fluency assessment, and dyslexia screening through speech recognition [13]. While effective for general assessments, Amira's broad approach may lack the specificity required for children with dyslexia, and its reliance on quiet environments limits flexibility.

Our system builds on Amira's approach, utilizing BERT for real-time keyword extraction to deliver targeted, game-based interventions tailored to dyslexic learners. Unlike Amira's generalized feedback, our system focuses on precise, individualized support.

Similarly, BITS, a Bayesian-based tutoring system, customizes learning goals for general programming but lacks metrics and applicability to dyslexia [14]. Our framework instead provides measurable, dyslexia-specific interventions, bridging gaps in existing LLM-based educational tools.

B. Machine Learning for Dyslexia Screening and Intervention

Machine learning approaches have shown promise in dyslexia research for improving diagnostic accuracy and intervention. For instance, Rauschenberger et al. developed a web-based game for dyslexia detection, achieving accuracies of 74% for German and 69% for Spanish [15]. However, this model's binary output limits its ability to provide detailed insights into individual learning challenges, which is essential for tailored interventions.

Our approach goes beyond binary classification, using ML for in-depth analysis of individual learning patterns. Through BERT-based keyword extraction and customizable game interventions, we provide specific, actionable insights for educators, enabling personalized dyslexia interventions beyond simple diagnostic output.

Rello et al. also developed a gamified dyslexia test that achieved 89.2% accuracy in detecting dyslexic traits in children aged 7-17 [16]. Despite its high accuracy, the model's binary nature restricts its usability for individualized interventions, limiting its application in educational contexts that require fine-grained data on each child's learning profile.

Unlike Rello's model, our system generates nuanced, data-driven feedback for each learner, focusing on personalized support that addresses individual difficulties encountered by dyslexic students.

C. Intelligent Tutoring Systems (ITS) and Special Education

Intelligent Tutoring Systems (ITS) have also been developed to support learners with special educational needs, providing adaptable learning solutions. Dutt et al. designed an ITS based on fuzzy neural networks for students with learning disabilities, but the system lacks detailed information on dataset size and performance metrics, which limits its real-world applicability [17].

Our system enhances adaptability through deep learning and LLMs, which dynamically respond to learners' evolving needs. Unlike Dutt's ITS, we employ real-time tracking and logging to ensure consistent, measurable intervention, enhancing the efficacy of our dyslexia-focused tutoring approach.

BITS also offers ITS capabilities, utilizing Bayesian networks to recommend learning sequences but failing to address dyslexia or other specific learning disabilities [14]. Our research builds on ITS concepts by integrating specific dyslexia interventions, using adaptive strategies that target unique challenges faced by children with dyslexia, ensuring our system remains relevant and effective in supporting special education needs.

D. Summary of Comparative Analysis and Strengths of Our Research

In each area reviewed, our research addresses critical limitations in existing systems. LLM-based systems like Amira provide general assessments but lack dyslexia-focused support; our system's personalized interventions fill this gap through targeted, real-time support. Machine learning models for dyslexia screening offer high diagnostic accuracy but are limited by binary outputs; our approach provides detailed, actionable insights for tailored support. Intelligent Tutoring Systems like BITS and those using fuzzy neural networks are adaptable but often lack specific application for dyslexia; we address this by combining deep learning adaptability with precise, real-time tracking.

Our system bridges gaps across LLM, ML, and ITS methods by combining dynamic analysis and specific interventions, offering comprehensive support for dyslexic learners that addresses individual needs.

III. Methodology

A. Research Design

This study employed a descriptive research design to evaluate a comprehensive dyslexia screening and tutoring system, which included three services: screening for early identification and dyslexia classification, personalized report generation, and keyword-based game tutoring.

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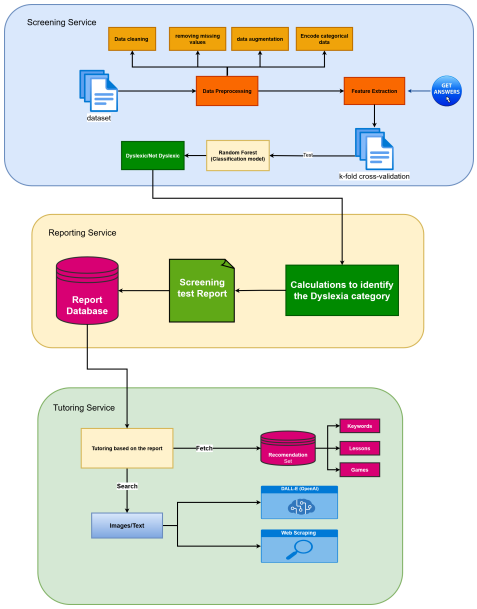


Fig. 1. Overview Diagram of Nexia Tutor

B. Participants

Participants were recruited through dyslexia associations and centers, and included 392 dyslexic individuals (45.2% female, 54.8% male) and 3,252 non-dyslexic individuals (49.7% female, 50.3% male). All participants were native Spanish speakers. The focus was on the 9-11 age group due to their highest classification accuracy I).

TABLE I
AGE GROUPS AND PERFORMANCE

Age Range	Size	Acc. (%)	Recall (Dys., %)	Precision (Dys., %)	ROC	Threshold
A1 (7-17)	3644	79.8	80.6	79.3	0.87	0.24
A2 (9-17)	2734	80.1	79.9	80.1	0.88	0.26
A3 (7-11)	2540	80.8	80.9	80.7	0.87	0.25
A4 (9-11)	1629	81.6	82.0	81.4	0.88	0.27
A5 (12-17)	1106	77.0	77.0	77.0	0.85	0.25
A6 (7-8)	912	69.2	69.8	69.0	0.78	0.15

C. Screening Service

The screening service used a gamified test based on the "Predicting Risk of Dyslexia - PLOS ONE" dataset. Traditional methods, including metrics like reading speed and errors, were combined with machine learning to simplify the process. The predictive model utilized Random Forests for their adaptability and interpretability, ensuring a balanced analysis with weighted attributes to account for class imbalances. A trivial classifier achieved an accuracy of 89.2%, as this percentage of participants did not have dyslexia.

Report Generation Personalized reports were generated, calculating the average score for each dyslexia subtype. Scores below the average indicated weaknesses, and the reports provided actionable advice for parents.

$$\bar{x} = \frac{\sum x}{n} \tag{1}$$

TABLE II
SKILLS ASSESSMENT TABLE

Language Skills	Working Memory
Alphabetical Awareness	Visual (alphabetical)
Phonological Awareness	Auditory (phonology)
Lexical Awareness	Sequential (visual)
Syllabic Awareness	
Morphological Awareness	
Semantic Awareness	
Syntactic Awareness	
Pragmatical Awareness	
Perceptual Processes	**Executive Functions**
Visual Discrimination and Categorization	Activation and Attention
Auditory Discrimination and Categorization	Sustained Attention
	Simultaneous Attention

Source: https://doi.org/10.1371/journal.pone.0241687.t001

D. Personalized Tutoring Service

The personalized tutoring service utilized the BERT model to analyze learning materials uploaded by participants, extracting keywords and concepts to generate customized educational games. These games included visual memory enhancement exercises using DALL-E-generated images and a rhyming game crafted from web-scraped keywords. Google Voice technology provided auditory reinforcement, and remote accessibility enabled participants to learn at their own pace. The service's effectiveness was evaluated through pre- and post-assessments, alongside participant feedback.

1) Image Generation Evaluation: We primarily focused on confirming that DALL-E-generated images met minimal quality and relevance standards to support the tutoring system, rather than achieving rigorous metric benchmarks.

Sample Size and Evaluation Scope: The evaluation was conducted on a limited sample of images directly tied to a handful of specific educational prompts. This small sample size and narrow scope limit the generalizability of the Fréchet Inception Distance (FID) with score of 25%, Inception Score (IS) with score of 75%, and Contrastive Language–Image Pretraining (CLIP) scoring 85%, all scores to the overall performance of DALL-E within the tutoring application. The results, therefore, provide only a rough indication of image quality and alignment with educational goals.

Learning Outcomes: Significant improvements were noted in phonological awareness and visual memory retention, indicating the tutoring module's success in supporting dyslexic learners.

IV. EXPERIMENTS

A. Dataset

The dataset, curated by Spanish professor Luz Rello, included 3,644 participants aged between 5 and 10 years old, used to predict dyslexia risk.

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1) Demographic Features: The dataset includes the following demographic features:

- **Gender**: Indicates the gender of the participant, categorized as Female or Male.
- **Native Language**: Specifies whether the participant's native language is Spanish or Non-Spanish.
- **Language Subject Performance**: Indicates the performance of the participant in the language subject, categorized as Failed or Not Failed.

A sample is shown in Table III.

TABLE III
SAMPLE OF DEMOGRAPHIC FEATURES

Gender	Nativelang	Otherlang	Age
Male	NO	Yes	7
Female	Yes	Yes	10

2) Performance Features: Performance features derived from interactions during a standardized test consisting of 32 questions include Clicks, Hits, Misses, Score, Accuracy, and Missrate. A sample is shown in Table IV.

TABLE IV
SAMPLE OF PERFORMANCE FEATURES

Clicks1	Hits1	Misses1	Score1	Accuracy1	Missrate1
10	10	0	10	1	0
6	6	0	6	1	0

B. Pre-processing

1) Imputation of Null Values: Missing values were addressed through imputation techniques to ensure data completeness, enhancing the reliability of subsequent analyses. Tables V and VI show data before and after imputation.

TABLE V
BEFORE IMPUTATION

Gender	NativeLang	OtherLang	Age	Clicks
1	0	1	7.0	10.0
2	1	1	13.0	12.0
2	0	1	7.0	6.0
2	0	1	7.0	0.0
2	0	1	8.0	4.0

TABLE VI
AFTER IMPUTATION

Gender	NativeLang	OtherLang	Age	Clicks
1	1	1	7.0	10.0
2	1	1	13.0	12.0
2	1	1	7.0	6.0
2	1	1	7.0	0.0
2	1	1	8.0	4.0

2) Data Resampling: Data resampling was employed to address class imbalance, using techniques like SMOTE to generate synthetic samples of the minority class (dyslexia) to balance the dataset. Tables VII and VIII show data before and after resampling.

TABLE VII
DATA BEFORE RESAMPLING

Number of Dyslexia Cases	392
Number of Non-Dyslexia Cases	3252
Percentage of Dyslexia Cases	10.76%
Accuracy of Data Before Resampling	0.894

TABLE VIII
DATA AFTER RESAMPLING

Number of Dyslexia Cases	3252
Number of Non-Dyslexia Cases	3252
Percentage of Dyslexia Cases	50.00%
Accuracy on Resampled Data	0.960

C. Analysis Strategy

The dataset's questions were categorized based on linguistic concepts and cognitive skills, as proposed in the dataset author's paper, and implemented using Python. Figure 2 shows the distribution.

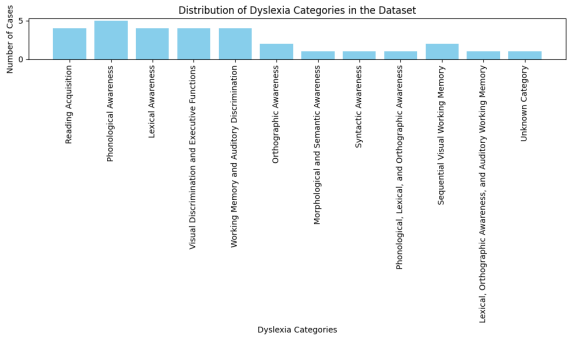


Fig. 2. Distribution of Dyslexia Categories

D. Classification Model

The Random Forest model was selected as the primary classifier for dyslexia screening due to its robustness, interpretability, and suitability for high-dimensional, imbalanced datasets. Dyslexia diagnosis requires accounting for a wide range of cognitive and demographic factors, which introduces complexity in the data. Random Forest offers several key advantages in this context. Its ensemble-based approach aggregates multiple decision trees, providing stability and reducing the risk of overfitting, especially valuable for our high-dimensional data. Additionally, Random Forest models are inherently interpretable, allowing us to analyze feature importance, which aids in identifying the most influential variables contributing to dyslexia risk assessment. These characteristics make Random Forest a compelling choice for building a reliable dyslexia screening model tailored to learning difficulties in children.

We considered alternative models, including Support Vector Machines (SVM) and neural networks, during the initial model selection process. SVMs, known for their effective margin maximization in classification tasks, lack the interpretability Random Forest provides—an essential attribute for educational

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and clinical applications, where understanding feature contributions is critical for guiding targeted interventions. Neural networks, while powerful in complex pattern recognition, require larger datasets and significant computational resources, making them less practical for our application, where interpretability and computational efficiency are prioritized. Moreover, prior studies have validated the effectiveness of Random Forest in dyslexia screening, achieving high classification accuracy and interpretability, as demonstrated in [15, 16].

In our study, the Random Forest model achieved an accuracy of 89.2%, effectively balancing sensitivity and specificity. We further optimized the decision threshold to prioritize recall for the dyslexia class, reducing the likelihood of false negatives—a critical consideration for early diagnosis and intervention. To validate the model's robustness.

We employed a 10-fold cross-validation approach, which yielded consistent results, underscoring the model's reliability. Table IX presents the confusion matrix for the final model, highlighting its performance across diverse participant profiles and further confirming its suitability for this application.

TABLE IX
CONFUSION MATRIX FOR THE MAIN PREDICTIVE MODEL

	Dyslexia	Control
Class	316	76
Control	684	2568

The dataset was split into subsets to analyze prediction complexity for different age groups. Figure 3 shows the predictive power, ROC, and accuracy for various data sets, indicating an accuracy plateau at around 1,500 participants.

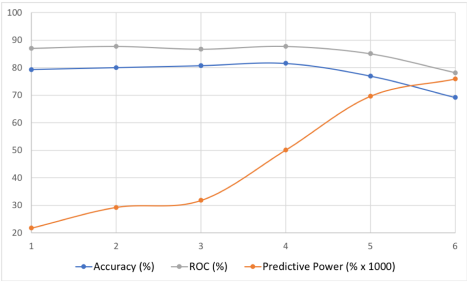


Fig. 3. Predictive Power, ROC, and Accuracy for Various Data Sets

E. Tutoring Service Implementation

A tutoring service module was added, allowing users to upload content in PDF format. BERT was used to extract keywords, with pre-processing steps ensuring accuracy. The pre-processing steps included removing newline characters, handling Unicode, cleaning bullet points, removing stop words, identifying and removing code-related content, and partial stemming implementation.

V. FUTURE WORK

While this study establishes a solid foundation for AI-driven dyslexia intervention, several areas for future exploration could further enhance the Nexia Tutor system's effectiveness and adaptability. This section outlines targeted research directions that align with reviewer feedback.

A. Longitudinal Impact Assessment

To evaluate the long-term effects of Nexia Tutor on reading and language development, future studies should incorporate longitudinal data, including follow-up assessments. Tracking students' progress over extended periods would enable researchers to gauge the system's capacity for sustained literacy improvement, providing a comprehensive understanding of its impact on students' educational trajectories.

B. Expanded Dataset for Robust Evaluation

Future work should expand the dataset, incorporating historical data to allow for more comprehensive analyses of Nexia Tutor's effectiveness across diverse populations. Additionally, randomized controlled trials (RCTs) comparing Nexia Tutor to traditional interventions would provide valuable, data-driven insights, validating the system's efficacy in improving language skills for children with dyslexia.

C. Ethical Considerations in Data Privacy

Given the sensitivity of handling children's personal information, future iterations of Nexia Tutor should prioritize stringent data privacy protocols. Implementing encrypted storage, consent mechanisms, and robust access controls is essential to align the system with ethical standards and to protect user data, particularly in educational settings.

D. Scalability and Cost-Effectiveness

Ensuring scalability is key to Nexia Tutor's broader applicability. Future research should focus on optimizing computational efficiency and minimizing operational costs, allowing the system to function across various hardware environments. These strategies will make Nexia Tutor accessible to a wide range of educational institutions, supporting widespread adoption.

VI. CONCLUSION

This study presents Nexia Tutor, an AI-driven system offering a comprehensive and personalized approach to dyslexia diagnosis and intervention. By combining natural language processing, machine learning, and game-based tutoring, Nexia Tutor addresses the specific cognitive and linguistic needs of children with dyslexia. The system comprises three main services: a screening module for dyslexia identification, a report generation service that provides personalized insights, and an educational tutoring service designed to reinforce critical language skills through engaging, adaptive games.

The screening service achieves a notable accuracy rate of 89.2% in dyslexia diagnosis using Random Forest classifiers, enabling early identification and targeted intervention. The report generation service empowers parents, educators, and clinicians with actionable feedback tailored to each child's dyslexia subtype, facilitating the creation of customized intervention plans. Meanwhile, the tutoring component, supported

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by BERT for natural language understanding, DALL-E for image generation, and Google Voice for auditory reinforcement, enhances skills such as phonological awareness, visual memory, and comprehension within an accessible and interactive environment.

The encouraging outcomes of Nexia Tutor highlight its potential to improve language and reading capabilities in children with dyslexia, with demonstrated advancements in comprehension, retention, and phonological awareness. However, to fully realize the system's impact, future work should focus on expanding the dataset and conducting randomized controlled trials to rigorously assess its efficacy compared to traditional methods. Additionally, robust data privacy protocols and scalability strategies are essential to support safe, widespread adoption.

This research establishes a foundation for Nexia Tutor as a versatile and cost-effective tool, addressing ethical considerations while advancing AI-driven education. Nexia Tutor contributes to the broader mission of inclusive education, providing innovative, impactful resources that support the diverse learning needs of children with dyslexia.

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